8th Grade Unit 9 Assessment Guide

General Information	Unit 9 is students' introduction to transformations. Students will further study transformations in Geometry. The benchmarks MA.8.GR.2.1 and MA.8.GR.2.2 are in the plane while MA.8.GR.2.3 is on the coordinate plane. The Big-M states that "single transformations include one vertical translation or one horizontal translation. A vertical and horizontal translation would be considered two transformations." If students struggle with rigid transformations, teachers may wish to use patty paper to represent the plane that a figure is on. By moving the patty paper, the student builds understanding that all of the points move. Teachers may also wish to use components from Geometry Unit 1 to help support students who struggle with transformations. This unit was written to bridge the year gap between Grade 8 and Geometry. The unit includes MA.8.GR.2.4 so that students understand the connection between similar triangles and dilations. Question 8 on this assessment revisits MA.8.GR.2.4. Teachers may wish to note if students struggle on question 8 but understand how to find scale factors given lengths in questions 4 and 5. Teachers may find that students struggle when the measurements are not provided on the figure. Consider monitoring student work for question 8b to determine if students do not recall the Pythagorean theorem from Unit 5. The writers of the curriculum looked for ways to spiral content. This allows teachers and students to revisit content. If most of your students struggled with 8b because of the Pythagorean theorem, consider reteaching it or adding a spiral review of the concept. Teachers also have the option to remove question 8b from the assessment, if desired, and adjusting the points as needed.
Calculator Information	The questions on the unit assessment are calculator neutral. Teachers can determine whether students should use a calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

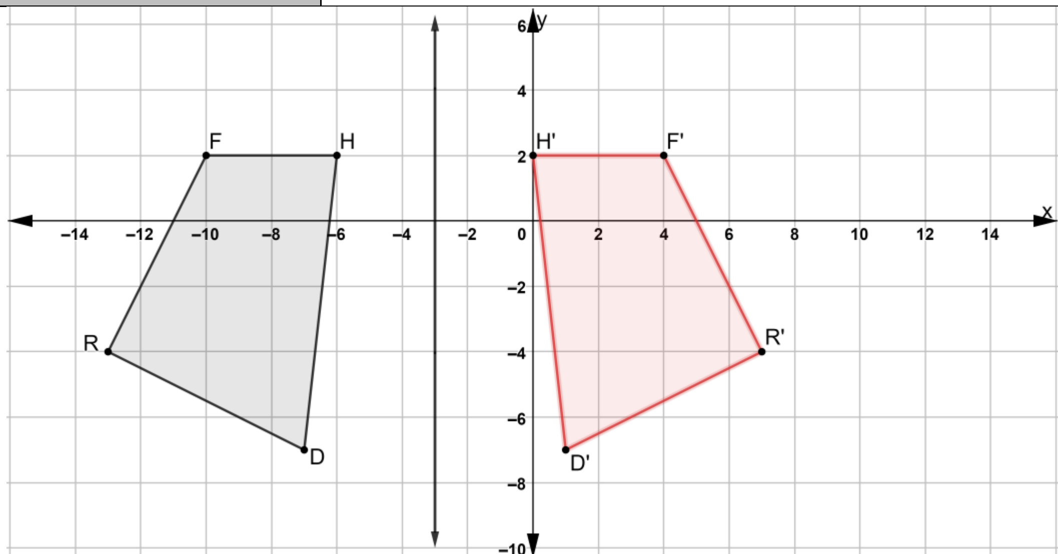
Question	**1**	**Benchmark(s)**	MA.8.GR.2.1
	☐ reflection ☐ rotation ☐ translation	**Recommended Scoring** **(5 Total Points)**	Consider not giving partial credit.
Question	**2**	**Benchmark(s)**	MA.8.GR.2.1
	☐ reflection ☐ rotation ☐ translation	**Recommended Scoring** **(5 Total Points)**	Consider not giving partial credit.
Question	**3**	**Benchmark(s)**	MA.8.GR.2.1
	☐ reflection ☐ rotation ☐ translation	**Recommended Scoring** **(5 Total Points)**	Consider not giving partial credit.
Question	**4**	**Benchmark(s)**	MA.8.GR.2.2
	4	**Recommended Scoring** **(5 Total Points)**	Consider not giving partial credit.
Question	**5**	**Benchmark(s)**	MA.8.GR.2.2
	0.6 or $\frac{3}{5}$	**Recommended Scoring** **(5 Total Points)**	Consider not giving partial credit.

Question	6a	Benchmark(s)	MA.8.GR.2.3

F'	H'	D'	R'
$(-10, 9)$	$(-6, 9)$	$(-7, 0)$	$(-13, 3)$

Recommended Scoring **(20 Total Points)**	Consider giving 5 points for each coordinate pair.

Question	6b	Benchmark(s)	MA.8.GR.2.3



Recommended Scoring **(10 Total Points)**	Consider giving 2 points for each correctly plotted coordinate pair and 2 points for correctly labeling the transformation.

Question	6c	Benchmark(s)	MA.8.GR.2.3

Transformation A is rotation of
- ○ 90° clockwise
- ○ 180° clockwise
- ○ 270° clockwise

about point W.

Transformation B is rotation of
- ○ 90° counterclockwise
- ○ 180° counterclockwise
- ○ 270° counterclockwise

about point W.

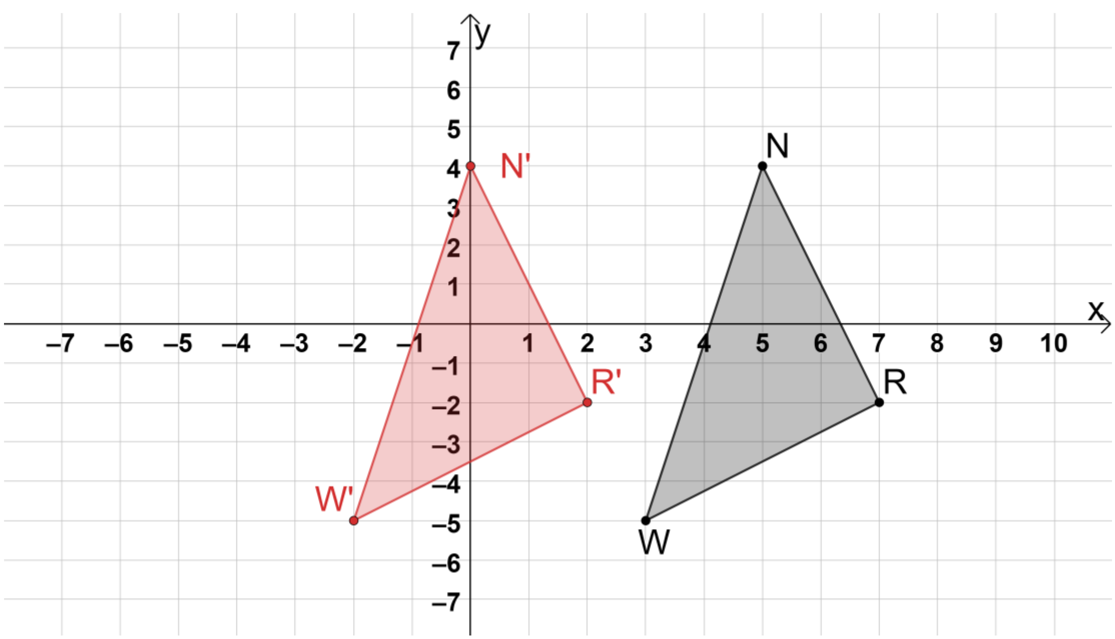
Recommended Scoring **(10 Total Points)**	Consider giving 5 points for each correct choice.

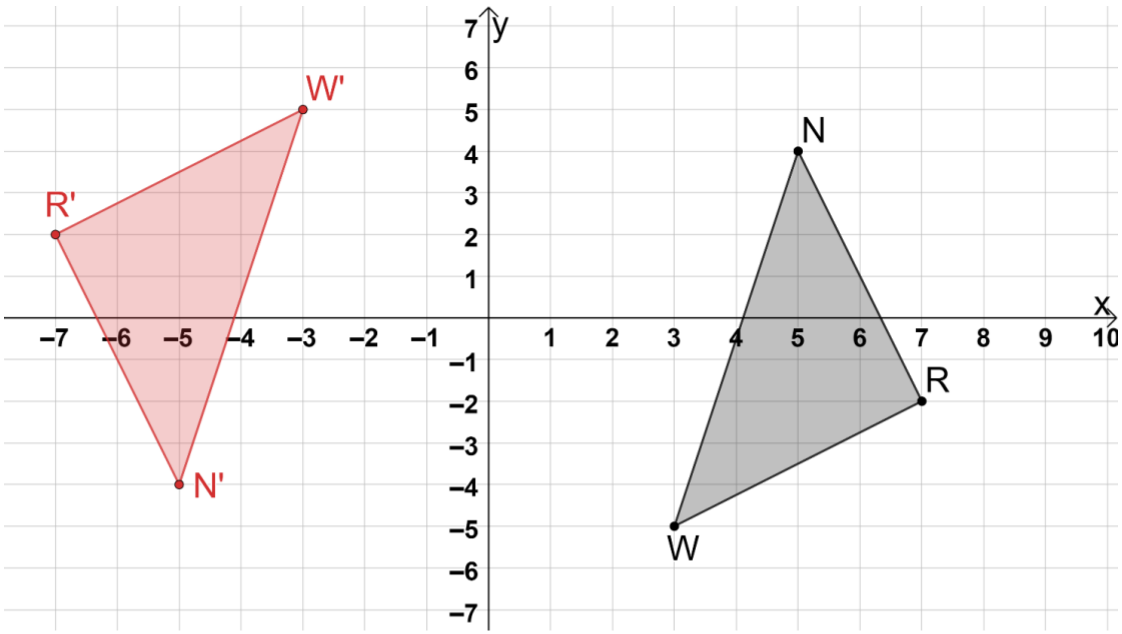
<table>
<tr><th>Question</th><th>7</th><th>Benchmark(s)</th><th>MA.8.GR.2.1, MA.8.GR.2.2</th></tr>
<tr><td colspan="2">When the mapping of a transformation of a figure shows that the
transformation
○ preserves congruence,
○ has corresponding vertices,
○ is a scaled copy of itself with a scale factor other than 1,
then the transformation is a
○ slide.
○ dilation.
○ rigid transformation.

or

When the mapping of a transformation of a figure shows that the
transformation
○ preserves congruence,
○ has corresponding vertices,
○ is a scaled copy of itself with a scale factor other than 1,
then the transformation is a
○ slide.
○ dilation.
○ rigid transformation.</td><td>Recommended Scoring

(5 Total Points)</td><td>Consider giving 5 points for each correct choice.</td></tr>
</table>

Question	8a	Benchmark(s)	MA.8.GR.2.4
12.5 ft		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**8b**	**Benchmark(s)**	MA.8.GR.2.4
20.01 ft		Recommended Scoring (5 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**9**	**Benchmark(s)**	MA.8.GR.2.3
		Recommended Scoring (10 Total Points)	Consider giving 3 points for each correctly plotted coordinate pair and 1 point for correctly labeling the transformation.

Question	10	Benchmark(s)	MA.8.GR.2.3
		**Recommended Scoring** **(10 Total Points)**	Consider giving 3 points for each correctly plotted coordinate pair and 1 point for correctly labeling the transformation.